

CUSTOM 3DS THEME WORKFLOW

A general, repeatable guide based on the tools and steps that actually worked

Goal: build a custom Nintendo 3DS theme with custom top/bottom artwork, colors, icon, and looping background music; package it; transfer it over Wi-Fi; test it on real hardware; and do a privacy pass before sharing.

1. Tools and What They Do

Tool	Job in the workflow
Photopea	Crop, resize, compose, and export screen art and optional theme icons as PNG.
Kame Editor	Build the visual theme, customize UI/colors, preview it, save the editable .rc project, and deploy the compiled design.
Internet Archive	One place to locate source audio/material. It was used to find the audio in this workflow.
Apple Music	GUI fallback for converting MP3 to WAV when another tool will not accept MP3.
Homebrew	macOS package manager used to install FFmpeg.
FFmpeg	Precisely trim/convert audio, including millisecond-level start times and durations.
mu-wave	Preview loops and create looping BCSTM audio for the 3DS.
Themezer Create & Share	Combine body_LZ.bin, bgm.bcstm, icon, and metadata into the final theme ZIP.
FTPD	Run an FTP server on the 3DS so its SD card can be accessed over local Wi-Fi.
Cyberduck	FTP client used on the computer to transfer the final ZIP to the 3DS.
Anemone3DS	Install, preview, and hardware-test themes on the 3DS.
sips (macOS)	Inspect PNG dimensions, alpha state, color profile, and basic image properties.

2. Step 1 - Prepare the Artwork

Create the top and bottom images in Photopea. For the Kame workflow, use a **512 x 256** canvas for each screen texture. Crop/position the artwork on that canvas and export a normal flattened PNG.

For a custom theme-list icon, make a separate PNG with both dimensions divisible by 2. A square such as **48 x 48** worked well.

Tip: editor previews are useful, but the physical 3DS is the final judge of cropping and readability.

3. Step 2 - Build the Visual Theme in Kame

- Create a new theme in Kame Editor.
- Load the top-screen and bottom-screen PNGs.
- Adjust the options you care about: palette/colors, cursor, folders, file frames, text, arrow buttons, Open/Close buttons, and other UI elements.
- Save the editable Kame project (.rc). Keep this as your master project for future edits.
- Preview the theme, but expect the real console to differ slightly.
- Deploy/export when ready. The important compiled visual file for the finishing workflow is **body_LZ.bin**.

4. Step 3 - Find and Prepare Audio

Use an audio source you are permitted to use/share. In the workflow that produced this guide, **Internet Archive** was used to locate the source audio:

<https://archive.org/>

If your source is MP3 and the next tool refuses it, convert it to WAV. Apple Music's WAV Encoder worked as a simple GUI fallback. A standard working format was 16-bit PCM, 44.1 kHz, stereo.

5. Step 4 - Install FFmpeg on macOS

If Homebrew is installed:

```
brew install ffmpeg
```

Confirm FFmpeg is available:

```
ffmpeg -version
```

6. Step 5 - Cut the Exact Music Loop

Use FFmpeg to make a short WAV containing only the section you want on the HOME Menu. General pattern:

```
ffmpeg -ss HH:MM:SS.mmm -i "/path/to/source.wav" -t SECONDS.mmm -c:a pcm_s16le  
"/path/to/output.wav"
```

Example: start 3 minutes, 14.600 seconds into a song and keep 15.830 seconds:

```
ffmpeg -ss 00:03:14.600 -i "/path/to/song.wav" -t 15.830 -c:a pcm_s16le "/path/to/theme-loop.wav"
```

You can tune both the start time and duration by thousandths of a second. This is useful when a loop is almost right but has a tiny audible cut.

7. Step 6 - Turn the WAV into a Looping BCSTM with mu-wave

- Open mu-wave and load the trimmed WAV.
- Select the appropriate channel count (normally 2ch for stereo).
- Enable the Loop checkbox.
- Use the preview controls to listen across the loop boundary and adjust if necessary.
- Choose **BCSTM (3DS)** as the output format. Do not accidentally choose BRSTM (Wii).
- Build the BCSTM.
- Save the generated file and name it **bgm.bcstm**.

Why mu-wave mattered: it gave us a usable loop preview and produced a BCSTM whose loop flag actually worked on the 3DS.

8. Step 7 - Finish the Package in Themezer

- Open Themezer's 3DS Create & Share page.
- Load **body_LZ.bin** as the theme design.
- Load **bgm.bcstm** as the background music.
- Enter the theme name and required theme/BGM information.
- Add the custom icon if desired.
- Keep the theme private while testing if you do not want it publicly listed yet.
- Download the resulting ZIP. This ZIP is what Anemone3DS will install.

Important: the Themezer workflow used here expects an actual **bgm.bcstm**. The WAV is converted to BCSTM before this step.

9. Step 8 - Transfer the ZIP with FTPD + Cyberduck

On the 3DS, launch FTPD and leave it running. FTPD displays the console's local IP address and port. On the computer, open Cyberduck and create an FTP connection using exactly the address/port shown by FTPD.

Navigate to the SD card's **/Themes/** directory and drag the finished theme ZIP into it. **Do not unzip the theme.** Wait for the upload to complete, then disconnect Cyberduck and close FTPD.

10. Step 9 - Install and Test with Anemone3DS

- Open Anemone3DS.
- Find the newly transferred theme in the theme list.
- Preview it if useful.
- Install the full theme including its BGM.
- Return to the HOME Menu.
- Check the top image, bottom image, colors/UI, theme icon, music, and especially the loop transition.
- If the loop has a click, gap, or ugly cutoff, go back to FFmpeg, adjust the start/duration slightly, rebuild the BCSTM, repack, and test again.

11. Step 10 - Privacy Check Before Sharing

For PNGs created from screenshots, export normal flattened/cropped images. Pixels removed by a normal destructive crop are not recoverable by simply 'uncropping' the final flattened PNG.

On macOS, inspect each actual source PNG used in the theme:

```
sips -g all "/path/to/image.png"
```

Normal fields include width, height, PNG format, DPI, RGB, bit depth, alpha state, and an sRGB color profile. **sRGB IEC61966-2.1 is a color profile, not a personal profile.**

Before a public upload, also inspect the final ZIP: verify its filenames, included preview/icon, theme metadata, and that no local computer paths or unintended personal information are present.

12. Common Problems We Actually Hit

Problem	What worked
MP3 is grayed out / cannot be selected	Convert it to a standard WAV first.
Kame audio controls are gray	Do not assume the source is invalid; Kame's audio UI was unreliable in this setup.
Need an exact loop point	Use FFmpeg for millisecond-level cuts, then preview/build the loop in mu-wave.
mu-wave gives .brstm	Change Output format to BCSTM (3DS), then rebuild.
Themezer says image dimensions must be divisible by 2	Resize the icon so both dimensions are even, e.g. 48 x 48.
Theme looks different from editor preview	Test on real hardware; treat the console as authoritative.
Need to update theme without removing SD card	Use FTPD on the 3DS + Cyberduck on the computer.

13. Short Version - Repeatable Pipeline

1. Photopea -> make screen PNGs/icon.
2. Kame -> build visuals, save .rc master, export body_LZ.bin.
3. Internet Archive/other permitted source -> obtain audio.
4. WAV conversion if needed.
5. FFmpeg -> trim exact loop segment.

6. mu-wave -> enable Loop -> BCSTM (3DS) -> save bgm.bcstm.
7. Themezer -> body_LZ.bin + bgm.bcstm + icon/info -> download ZIP.
8. FTPD + Cyberduck -> upload ZIP to /Themes/.
9. Anemone3DS -> install and hardware-test.
10. sips + final ZIP inspection -> privacy pass before sharing.

This guide is intentionally general: keep your own theme names, usernames, file paths, and copyrighted/source-material decisions separate from the reusable workflow.